



What is Difference between assertEquals() and assertTrue() in TestNG?

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When writing unit tests in Java, TestNG is a popular tool with different methods to check if things work as expected. Two of the most common methods are `assertEquals()` and `assertTrue()`. Even though they both check results, they are used in different ways. This article will explain how `assertEquals()` and `assertTrue()` are different.

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What is assertEquals()?

`assertEquals()` in [testNG](#) is an assertion method that checks whether two values are equal. It takes two parameters: the expected value and the actual value. If the two values are not equal, the test will fail, and an `AssertionError` will be thrown.

Syntax of assertEquals()

Syntax of assertEquals is: assertEquals(Object expected, Object actual)

Example of assertEquals()

Let's consider a simple Calculator class that has an add method:

```
import org.testng.Assert;
import org.testng.annotations.Test;

public class CalculatorTest {

    @Test
    public void testAdd() {
        Calculator calculator = new Calculator();
        int result = calculator.add(2, 3);

        // Using Assert.assertEquals to check if the result is 5
        Assert.assertEquals(result, 5, "Test failed: result is
not 5");

        System.out.println("Test passed: result is 5");
    }
}

class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

Output

Test passed: result is 5

What is assertTrue()?

assertTrue() is used to verify that a condition is true. It takes a single parameter: a boolean value indicating whether the condition is true or false. If the condition is false, the test will fail. It can take one or two parameters.

Syntax of assertTrue()

1. Single Parameter: A boolean expression that you expect to be true. If the expression is false, the test will fail.

```
Assert.assertTrue(condition);
```

2. Two Parameters: A boolean expression and a custom failure message. If the expression is false, the test will fail, and the custom message will be displayed.

If the condition is false, TestNG will throw an `AssertionError`, causing the test to fail, and the optional custom message will be shown in the test report.

Example of assertTrue()

Here's an example using `Assert.assertTrue()` in TestNG:

```
import org.testng.Assert;
import org.testng.annotations.Test;

public class CalculatorTest {

    @Test
    public void testPositiveResult() {
        Calculator calculator = new Calculator();
        int result = calculator.add(2, 3);

        // Using Assert.assertTrue to verify that the result is
        positive
        Assert.assertTrue(result > 0, "Test failed: The result
is not positive");

        System.out.println("Test passed: The result is
positive");
    }
}
```

```
}

@Test
public void testIsFive() {
    Calculator calculator = new Calculator();
    int result = calculator.add(2, 3);

    // Using Assert.assertTrue to verify that the result is
    equal to 5
    Assert.assertTrue(result == 5, "Test failed: The result
is not 5");

    System.out.println("Test passed: The result is 5");
}
}

class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
}
```

Output

Test passed: The result is positive

Test passed: The result is 5

Differences between assertEquals() and assertTrue() in TestNG

Feature	assertEquals()	assertTrue()
Purpose	Compares two values for equality.	Verifies that a condition is true.

Feature	assertEquals()	assertTrue()
Parameters	Two parameters: expected value and actual value.	Single parameter: a boolean condition.
Error Handling	Fails if the two values are not equ	Fails if the condition is false.
Use Cases	Suitable for comparing strings, numbers, objects, etc.	Suitable for validating boolean expressions or conditions.

Examples in TestNG

Here's how you would use these assertions in a TestNG test:

```
import org.testng.annotations.Test;
import static org.testng.Assert.assertEquals;
import static org.testng.Assert.assertTrue;

public class MyTest {

    @Test
    public void testAdd() {
        Calculator calculator = new Calculator();
        int result = calculator.add(2, 3);
        assertEquals(result, 5);
    }

    @Test
    public void testIsValid() {
        Validator validator = new Validator();
        boolean result = validator.isValid("hello");
        assertTrue(result);
    }
}
```

```
class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
}  
  
class Validator {  
    public boolean isValid(String input) {  
        return input != null && !input.isEmpty();  
    }  
}
```

Output:

```
[TestNG] Running:  
  /Users/user/Documents/MyTest.java  
  
PASSED: testAdd  
PASSED: testIsValid  
  
=====  
Default test  
Tests run: 2, Failures: 0, Skips: 0  
=====
```

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